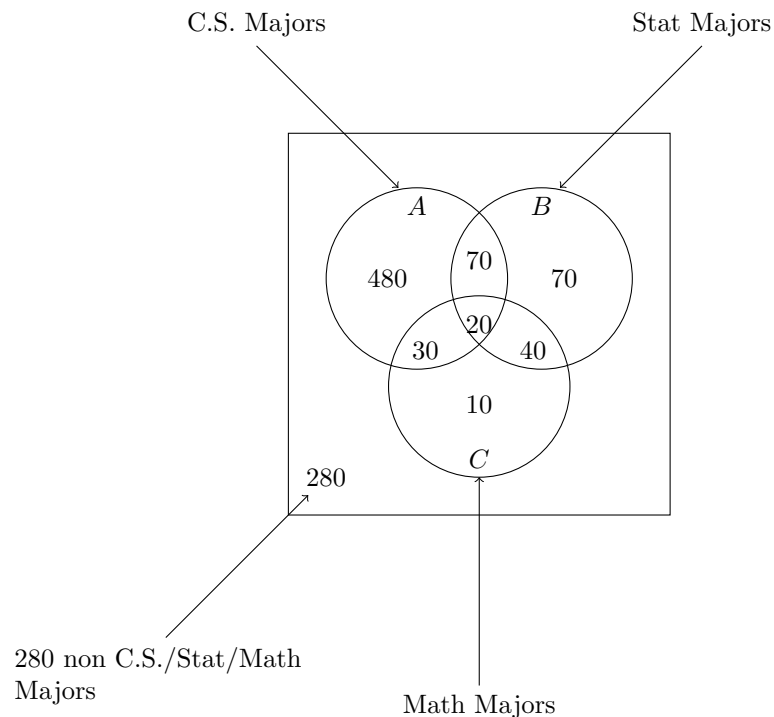


- 1:** Suppose there are 1000 students living in a certain dorm on Busch campus, of whom
 600 are CS majors,
 200 are Stat majors,
 100 are Math majors,
 90 are CS & Stat double majors (double majors are included in the number of single majors)
 50 are CS & Math double majors
 60 are Stat & Math double majors, and
 20 are CS & Stat & Math triple majors (these are also counted as double and single majors). What is the probability that a randomly selected student living in this dorm is...
- (a) ...majoring in at least one of CS, Stat or Math?
- (b) ...majoring in Math or double majoring in CS & Stat?

Solution: We start by making a venn diagram of the entire space



- (a) Now we can clearly see that there are 720 total math/c.s./stats majors in the dorm so

$$P(\text{student is a math/c.s./stats major}) = \frac{720}{1000} = \frac{72}{100} = \boxed{0.72 = 72\%}$$

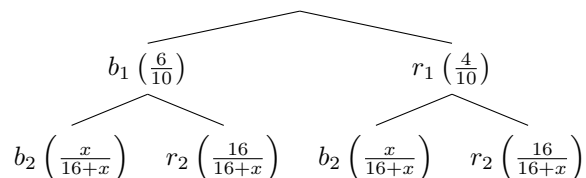
- (b) Using the venn diagram we have

$$\begin{aligned}
 &P(\text{student is a math major, or a CS and Stat double major}) \\
 &= P(\text{student is a math major}) + P(\text{student is a CS and Stat double major}) \\
 &\quad - P(\text{student is a math major and a CS and Stat double major}) \\
 &= \frac{100}{1000} + \frac{90}{1000} - \frac{20}{1000} \\
 &= \frac{17}{100} \\
 &= \boxed{0.17 = 17\%}
 \end{aligned}$$

2: An urn contains 10 balls: 4 red and 6 blue. A second urn contains 16 red balls and an unknown number of blue balls. A single ball is drawn from each urn. The probability that both balls are the same color is 0.44. Calculate the number of blue balls in the second urn.

A. 4 B. 20 C. 24 D. 44 E. 64

Solution: Let x be the number of blue balls in the other urn. We'll have b stand for blue and r for red, while the subscript 1 or 2 will indicate whether that color ball is drawn on the first or second turn. Without loss of generality we assume the urn with 10 balls is drawn from first.



Since the total probability of drawing two balls of the same color is $P = 0.44$ we can set the following equation and solve for x :

$$\begin{aligned}
 & \left(\frac{6}{10} \cdot \frac{x}{16+x} \right) + \left(\frac{4}{10} \cdot \frac{16}{16+x} \right) = 0.44 \\
 \Rightarrow & \frac{6x}{160+10x} + \frac{64}{160+10x} = 0.44 \\
 \Rightarrow & \frac{64+6x}{160+10x} = 0.44 \\
 \Rightarrow & 64+6x = 4.4x + 70.4 \\
 \Rightarrow & 1.6x = 6.4 \\
 \Rightarrow & \boxed{x = 4}
 \end{aligned}$$

and thus $\boxed{A}$ is the correct answer.

⁰L^AT_EX code for this document can be found on github [here](#)